

ALPACARUNS

Autonomous MoE Multi-Agent Trading on Alpaca

Live demo · Real trades · Real P&L

Two cooperating inference paths — an LLM agent graph
and a 6-expert MoE ensemble — funneled through one Go risk gate.

Paper-only. Fail-closed. No silent trades.

Welcome — splash + preview

The screenshot shows the Alpacaruns Demo Console interface. At the top, there's a navigation bar with links: Welcome, Live, Trades, Brain, Controls. Below the navigation bar, the status is "ALPACARUNS" with a green "ONLINE" indicator. A badge indicates "API REACHABLE · 80% OK" and "DEMO DASHBOARD · v0.0.4".

An autonomous multi-agent paper-trading bot. Live. Auditable. Code-enforced.

Alpacaruns runs two cooperating inference paths — a Google-ADK LLM agent graph and a deterministic mixture-of-experts ensemble — and funnels every order through the same Go-coded risk gate. This dashboard reads the live /JSON/ journal, the Alpaca account endpoint, and the decision log in real time.

Buttons: [Enter Live Dashboard](#) and [View Source on GitHub](#)

STACK

Go 1.20 · React 18 · Tailwind v3

MODES

LIVE · PAPER · AUTO · MANUAL

VENUE

Alpaca paper

SOURCE OF TRUTH

data/trades.json

01 - LIVE

Status & P&L

Equity current, drawdown, funds, agent positions

02 - TRADES

Explainability

Trade confidence, factor scores

03 - BRAIN

Agent views

Reasoning, signals, decision, feed

04 - CONTROLS

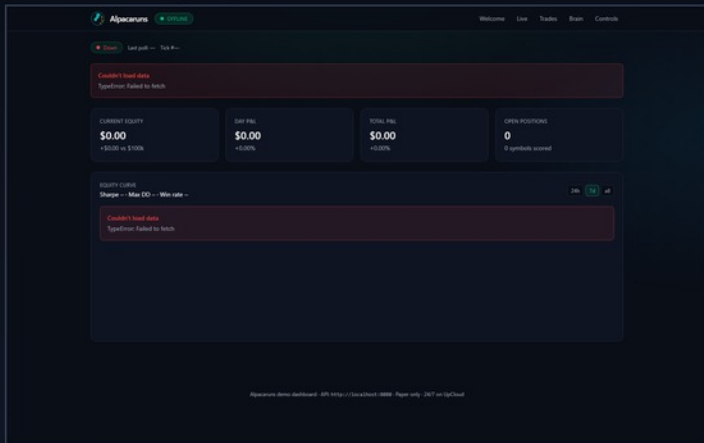
Safe actions

Pause / resume / stop

This UI is the eyes from the backoffice level — live status, trade log with explainability, an "agent brain" view, and most importantly, the controls. The data on the next pages is the real /JSON/ journal of the production bot running 24/7 on the UpCloud VM.

Alpacaruns demo dashboard · API: <http://localhost:8080> · Paper only · 24/7 on UpCloud

Live — status, P&L, equity curve



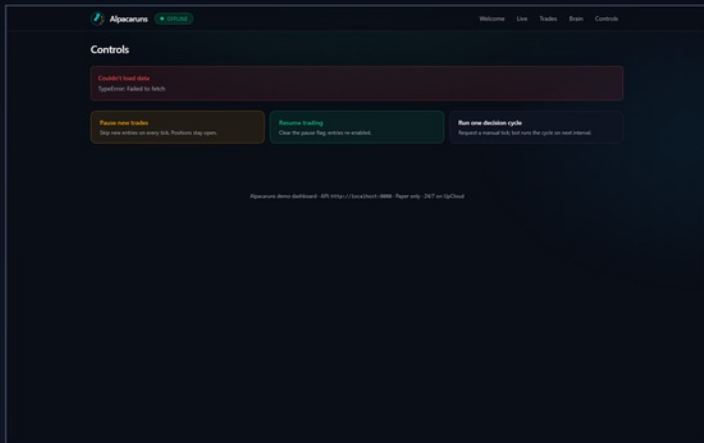
Trades — explainability

The screenshot shows the AlpacaRUNS Demo Console interface. At the top, there's a navigation bar with 'Welcome', 'Live', 'Trades', 'Bran', and 'Controls'. Below this is the 'Trade Log' section. It features a filter bar with 'All symbols' (dropdown), 'ALL', 'BUY', 'SELL', 'ALL', 'AGENT', 'ENSEMBLE', 'MANUAL', and a date range 'mm/dd/yyyy' with a calendar icon. Below the filter bar is a table with columns: TIME, SYMBOL, SIDE, QTY, PRICE, PATH, CORR., and NOTIONAL. The table is currently empty, displaying a red error message: 'Couldn't load data' and 'TypeError: Failed to fetch'. At the bottom of the interface, a footer note reads: 'Alpacaruns demo dashboard: API: <http://localhost:8080> - Paper only - 24/7 on UpCloud

Brain — agent + decisions

The screenshot displays the 'Agent Brain' section of the AlpacaRUNS demo console. At the top, the header includes the 'AlpacaRUNS' logo with a 'LIVE' status indicator and a navigation menu with links for 'Welcome', 'Live', 'Trades', 'Brain', and 'Controls'. The 'Agent Brain' title is prominently displayed. Below this, two identical error messages are shown in a light blue box: 'Couldn't load data' followed by 'TypeORM: failed to fetch'. At the bottom of the interface, a footer note states: 'AlpacaRUNS demo dashboard - API: <http://localhost:3000> - Paper only - DWT on UpCloud

Controls — safe actions



Three paths, one risk gate

LLM Agent

Google ADK multi-agent graph:
GatingRoot → TradingCycle → ExecutionEngine

MoE Ensemble

6 experts (trend, meanrev,
breakout, pairs, xsmom, seasonality)
+ performance-weighted gater

Deterministic auto

Pure factor pipeline:
factors.Engine → strategy.Decide
→ risk.Validator → Alpaca MCP

↓ **risk.Validator.Validate** ↓

fail-closed

Honest limitations + what to watch

- **Options overlay degrades**

/v2/options/snapshots 404s on free plan → bot logs skip + trades the equity path.

- **Local-LLM tool calling**

Qwen3 with --jinja chat template; malformed tool calls remain the #1 failure mode.

- **No backtester**

Thresholds validated by unit tests + live paper runs, not historical replay.

- **All paper only**

Fresh post-2026-08-28 paper account — no live capital at risk.